Havalyn Hale

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EXPERIENCE

Olin Electric Motorsports September 2026 - Present
Low Voltage Engineer *Needham, MA*

- Finishing the Formula SAE car's telemetry system: Python on a Raspberry Pi (Linux) relaying CAN bus data to a ground station over 915 MHz SiK radios
- Reconfigured the radios from 9600 to 57600 baud and redesigned the message format, cutting each packet from ~1,870 bytes to ~40 and raising throughput from ~3 to 100+ messages/sec; added per-ID rate limiting
- Link test at 151 ft, no line of sight: 4.8% loss over 500 messages (target <10%)
- Designing the ground station for a ~700 m track; built a Python link budget (free-space and two-ray ground reflection) predicting a 29.6 dB margin at 1 km against a 20 dB target

Olin Vine Robotics August 2026 - Present
Research Assistant (Independent Study) *Needham, MA*

- Designed a 6-channel steering control board (regulator, pressure sensor and vent valve per channel) for 6 DOF; building an expandable air manifold (2 DOF now, path to 5+)
- Built a Raspberry Pi Pico board reading a Honeywell ABP sensor's raw 14-bit output over I²C; wrote C firmware that rejects bad readings and zeroes at startup, cutting noise to a few thousandths of a psi
- Building a second board that reads 6 ABP sensors through a multiplexer across three robot sections

Olin Plasma Engineering Lab (OPEL) January 2026 - Present
Researcher *Needham, MA*

- Co-developed MATLAB models for the MK5 Hall Effect Thruster on an 8 kg CubeSat: sized propellant (0.11-0.22 kg for a 280 m/s apogee raise) and predicted ~15 mN thrust, 1,630 s Isp and 305 W; compiled results as an Olin ISR
- Rebuilt the MK4 Hall Effect Thruster in COMSOL to simulate its electromagnetic fields for a redesign
- Co-developed design requirements for a 3U CubeSat-scale thruster; performed thin-wall stress analysis of an Al 6061-T6 propellant tank (5 cm radius, 3 mm wall, 14.3 MPa), yielding a 1.08 safety factor

Olin Shop August 2026 - Present
Electronics Assistant *Needham, MA*

- Redesigned soldering training to add de-soldering; covers component ID, joint technique, safety, and schematic/BOM reading
- 3D printed cases for 2 Saleae logic analyzers (SOLIDWORKS); re-drew the training schematic in KiCAD

PROJECTS

Rocky Animatronic (PIE, team project)
12-servo walking robot for a Rocky (Project Hail Mary) animatronic; wired the servo driver, writing the C++ PWM gait, and diagnosing current draw and servo stall issues
C++, Arduino Nano

Introduction to Radio Frequency Parameters (QEA, self-selected topic)
Derived relationships between Z, Y, ABCD and S-parameters (including cascaded ABCD); used Smith Charts for impedance matching with complex port impedances; presented to 50+ students

Guitar Tuner: Analog Bandpass Filter Design
KY-037 sound sensor and cascaded RC/CR bandpass filter (190-450 Hz) that correctly identified guitar notes

EDUCATION

B.S. in Electrical and Computer Engineering *Olin College of Engineering* Expected May 2029
Needham, MA
4-Year 50% Merit Tuition Scholarship. GPA: 4.0. Relevant Coursework: Software Design, Discrete Mathematics (in progress), Modeling and Simulation, Quantitative Engineering Analysis, Instrumentation Sensors and Measurement

SKILLS

Programming: Python (NumPy), C, C++, MATLAB, Java
Embedded & Systems: Linux/Ubuntu, Raspberry Pi, CAN Bus, Serial Communication (pyserial), I²C, PWM, Arduino, RF Link Budgets
Tools & Lab: GitHub, Altium, KiCAD, SOLIDWORKS, COMSOL, Oscilloscope, Logic Analyzer, Soldering, 3D Printing